Conditional Coordinate Sweeps and Analytic Transfer

Explicit Line Gaps, Green Barriers, and Quantitative Endpoint Budgets

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M476

October 9, 2026

Abstract

We study the quantitative analytic interface in the September 26, 2026 manuscript *Conditional coordinate sweeps and analytic transfer*. The source already distinguishes conditioned rational laws from an unconditioned matrix polynomial, and already supplies a dimension-uniform complex majorant; it does not rely on the identity theorem alone. We derive an explicit, representation-independent contraction deficit for a binary line sweep using cube-edge word length and projection energy. This gives a common complex disk without enumerating irreducible representations. A disk Green potential then transfers a real-interval Schatten saving to the binary endpoint with an explicit integral exponent. Unlike the source's elementary logarithmic barrier, this potential permits a slit independent of the disk's excess radius and has a linear small-radius asymptotic. We give a dimensional-majorant budget, conditional pole and feasibility examples, and an explicit reciprocal-dimension remainder. The feasible-path moment theorem, signed-density estimates and simultaneous induction remain imported inputs, not consequences of these local certificates. Contribution assessment: C3/5, explicit line-gap and fixed-slit Green-barrier bounds provide a notable quantitative refinement of the source's analytic interface, without independently proving the conditional moment theorem or a new mixing-order result.

1 Problem, law, and scope

Let $N = 2^d$. A binary coordinate sweep acts on the slots $\{0,1\}^d$: in each coordinate direction, independently swap or fix each of its disjoint cube edges with probability $1/2$, and apply the directions in their specified order. Its law on S_N is B_N. Products gh apply h first. For a unitary representation ρ, the average is $K_\rho = \mathbb{E}_{B_N}\rho(g)$, with operator norm on its actual representation Hilbert space. The Schatten convention is $\|A\|_p^p = \mathrm{Tr}((A^*A)^{p/2})$, without division by dimension. All total-variation distances use half the L^1 distance.

The source [1] claims a universal power $\|K_\lambda\|_{\mathrm{op}} \le D_\lambda^{-g}$, $g > 0$, in every irreducible representation of S_N, and hence mixing after a fixed number of sweeps. One sweep is d physical Thorp shuffles, not one physical shuffle. The source's proof first changes the line law on larger coordinate blocks, proves a simultaneous conditioned estimate near the uniform line law, and transfers only the unconditioned average to the binary parameter.

This report establishes local statements in that chain. The explicit group comparison and Green-barrier arguments below are proved here. Classical word comparison, projection energy, subharmonic comparison, and finite-group Plancherel are not claimed as new principles. Reading

the entire cached manuscript is not an independent certification of its global conditioned induction. In particular, a finite representation test cannot certify the uniformity over all grids, all feasible trajectories, all Young diagrams, and all multiplicities required there.

2 Prescribed paths and the actual analytic object

Write a grid as $\Omega = \prod_{j=1}^{b} \Omega_j$, with $m_j = |\Omega_j|$ powers of two in a fixed interval $[R, R^2]$. On a line of size m, put

$$L_m(z) = (1-z)U_m + zB_m, \tag{1}$$

where U_m is uniform on S_m. This is a probability law for real $0 \le z \le 1$; complex z defines an operator-valued polynomial, not a probability. Each of U_m, B_m sends one specified card to a uniform slot and has zero mean sign when $m \ge 2$.

The coordinate path from x to y has position $(y_1, \dots, y_j, x_{j+1}, \dots, x_b)$ after stage j. A family H is feasible when these positions are distinct at every layer. Conditioning imposes an injective partial assignment on each used line; independence between original line variables makes the conditional law a product of the separate conditioned line laws. Feasibility in this sense is not positivity of that event at $z = 1$.

Proposition 2.1 (Line denominators)**.** *Suppose an event E prescribes u distinct inputs to u distinct outputs on a line of size m, $0 \le u \le m$. With $(m)_u = m(m-1)\cdots(m-u+1)$, set*

$$b_E = (m)_u B_m(E).$$

Then

$$L_m(z)(E) = \frac{1 + z(b_E - 1)}{(m)_u}. \tag{2}$$

Every conditioned matrix coefficient is rational in z, with possible pole at $-1/(b_E - 1)$ if $b_E \ne 1$. Cancellation may remove that pole, but does not make conditioning on a zero-probability physical event defined.

Proof. There are $(m-u)!$ permutations extending the partial injection, so $U_m(E) = 1/(m)_u$. Substitute in (1) and divide the unnormalized event-restricted matrix average by (2). Both numerator and denominator are affine; the assertion follows. □

For fixed R, only finitely many line sizes and partial assignments occur. Thus a sufficiently small common disk about zero avoids all these denominator zeros, independently of the number of lines. This observation alone gives no norm majorant uniform in the number of conditioned lines: products can magnify bounds on individual factors. It also does not extend the physical conditional laws to every real parameter up to one.

Example 2.2 (A genuine pole and an impossible binary event)**.** On four slots $0, 1, 2, 3$, first switch $(0,1), (2,3)$, then $(0,2), (1,3)$. The event $E = \{0 \mapsto 0,\ 3 \mapsto 3\}$ has uniform probability $1/12$, but binary probability $1/16$. The binary event forces all four switches to be off. The two uniform completions are the identity and $(1,2)$, equally likely. Thus

$$\Pr_{L_4(z)}\big((1,2) \mid E\big) = \frac{2(1-z)}{4-z}.$$

It has a nonremovable pole at $z = 4$, so even this finite conditional law is not entire. In contrast, $E' = \{0 \mapsto 0,\ 1 \mapsto 2\}$ is a feasible injective assignment for a uniform line but impossible for this

binary sweep: after the first layer both paths would require intermediate slot zero. The conditional uniform formula for $z < 1$ can have a removable limit, but there is no binary conditional probability given E'.

The source correctly avoids these difficulties in its endpoint argument: for $H = \varnothing$, the matrix $K_\lambda(z)$ is an unconditioned product of affine line averages, hence an entire matrix polynomial. Conditioning is used only in the real near-uniform theorem that supplies its small real-interval bound.

2.1 The imported simultaneous estimate

For $s = |\Omega|$, let $h = |H|$, $M = s - h$, and let λ be a partition of the free slots. The compatibility potential is

$$\mathcal{C}(H) = \sum_L \phi_{m_L}(h_L), \qquad \phi_m(u) = \log \frac{m^u}{(m)_u}, \qquad 0 \le \phi_m(u) \le u.$$

The source chooses fixed $R, q, z_* > 0$, with q an integer, and functions $c(s) \ge c_0 > 0, e(s)$, so that for every allowed grid, every feasible H, and every λ,

$$\log \|K_{\lambda,H}(z)\|_{2q}^{2q} \le -c(s) \log D_\lambda + e(s) h \log s - \mathcal{C}(H), \qquad 0 \le z \le z_*. \tag{3}$$

Here the conditional operators use unitary identifications between the different free-slot sets. Their singular values are the relevant quantities. We import (3), not prove it from a matrix fixture. With $H = \varnothing$ it gives exactly

$$\|K_\lambda(t)\|_{\mathrm{op}} \le D_\lambda^{-a_0}, \qquad a_0 = c_0/(2q), \quad 0 \le t \le z_*. \tag{4}$$

This extraction uses the unnormalized Schatten norm and introduces no dimension factor.

The potential cannot simply be discarded inside the induction. If t additional prescribed paths extend H to H^+, the source's placement cap is

$$p_H(x, y) \le s^{-t} \exp\{\mathcal{C}(H^+) - \mathcal{C}(H) + (\log 4) t b\}.$$

The increment cancels the child conditioned-moment potential at the appropriate fractional power. A statement only for typical paths or only for fixed representations would not justify that use.

3 A common disk from an explicit line gap

The source obtains a line contraction constant by taking a maximum over finitely many irreducibles and line sizes. The following constructive alternative works in every finite-dimensional unitary representation, including arbitrary multiplicities.

Theorem 3.1 (Cube-word projection deficit)**.** *Let $m = 2^b$, $b \ge 1$, and let $\rho : S_m \to U(\mathcal{H})$. Let A_i be the average over all independent switches in cube direction i, $Q = A_b \cdots A_1$, and $P = |S_m|^{-1} \sum_{g \in S_m} \rho(g)$. Define*

$$L_m = (m-1)(2b-1), \qquad \chi_m = \frac{1}{2bL_m^2}.$$

Then

$$PQ = QP = P, \qquad \|Q|_{\ker P}\|_{\mathrm{op}}^2 \le 1 - \chi_m. \tag{5}$$

Proof. Each switch layer is a subgroup, so A_i is an orthogonal projection. For a unit vector $v \in \ker P$, set $v_0 = v$, $v_i = A_i v_{i-1}$, and $e_i = \|v_i - v_{i-1}\|$. Orthogonality gives the exact telescoping identity

$$D := 1 - \|Qv\|^2 = \sum_{i=1}^{b} e_i^2.$$

If τ is one edge transposition in direction i, it fixes $A_i v$. Therefore

$$\|v - \rho(\tau)v\| \le 2\,\|(I - A_i)v\| \le 2\,(\|v - v_{i-1}\| + e_i) \le 2\sum_{j \le i} e_j \le 2\sqrt{bD}.$$

An arbitrary transposition of two cube vertices at graph distance a is a product of $2a-1$ edge transpositions along a shortest path: go to the last edge and back through the preceding edges. As $a \le b$, the length is at most $2b-1$. Cycle decomposition expresses any permutation as at most $m-1$ arbitrary transpositions. It consequently has an edge word of length at most L_m. Telescoping its unitary product yields $\|v - \rho(g)v\| \le 2L_m\sqrt{bD}$. On the other hand,

$$\frac{1}{m!}\sum_{g} \|v - \rho(g)v\|^2 = 2 - 2\,\mathrm{Re}\langle v, Pv\rangle = 2.$$

Thus $2 \le 4bL_m^2 D$, proving the norm bound. Uniform averaging absorbs every group element on either side, giving $PQ = QP = P$. No commutation of distinct coordinate projections is used. □

Corollary 3.2 (Explicit common radius)**.** *Fix $R = 2^\ell$, $\ell \ge 1$, and allow line sizes among the powers of two in $[R, R^2]$. Put*

$$B = 2\ell, \qquad \chi = \frac{1}{2B(R^2-1)^2(2B-1)^2}, \qquad r = 1 + \chi/4. \tag{6}$$

Every unconditioned grid average in any unitary representation satisfies $\|K(z)\|_{\mathrm{op}} \le 1$ for $|z| \le r$.

Proof. The denominator in χ_m increases with the line size, so $\chi_m \ge \chi$. Relative to $\operatorname{ran} P \oplus \ker P$, uniform averaging and (5) give

$$(1-z)P + zQ = I_{\operatorname{ran} P} \oplus zQ|_{\ker P}.$$

Since $0 < \chi \le 1/2$,

$$(1 + \chi/4)^2(1 - \chi) = 1 - \chi/2 - 7\chi^2/16 - \chi^3/16 < 1.$$

Every line average is contractive on the asserted disk. Independence gives ordered products for stage and sweep averages, and submultiplicativity proves the conclusion. □

For $m = 4$, the deficit is $1/324$; for $m = 2$, the actual deficit is one although the bound gives only $1/2$. The comparison is intentionally coarse. For an interval $[2, 4]$, formula (6) gives $r = 1 + 1/1296$. No practical mixing constant is inferred: R and the available near-uniform interval are fixed by the much deeper source induction. The benefit is an explicit dimension-free disk, not a measured algorithmic acceleration.

4 A fixed-slit Green barrier

We now replace only the barrier in the source's analytic step. Let

$$r > 1, \qquad 0 < u < v < 1, \qquad \ell_0 = v - u, \qquad \Omega_r = \{z : |z| < r\} \setminus [u, v].$$

The symbol ℓ_0 is a slit length, distinct from the bit-block parameter $\ell = \log_2 R$. Define

$$G_r(z,t) = \log \left| \frac{r^2 - tz}{r(z-t)} \right|, \tag{7}$$

$$U_r(z) = \int_u^v G_r(z,t)\, dt, \qquad M_r = \ell_0[1 + \log(2r/\ell_0)], \qquad h_r = U_r/M_r. \tag{8}$$

Lemma 4.1 (Boundary and slit normalization)**.** *The function h_r is harmonic on Ω_r, extends continuously to its closure, is zero on $|z| = r$, and satisfies $0 \le h_r \le 1$. At the binary point,*

$$\eta_r := h_r(1) = \frac{J_r(u,v)}{M_r} > 0, \qquad J_r(u,v) = \int_u^v \log \frac{r^2 - t}{r(1-t)}\, dt. \tag{9}$$

Proof. The numerator zero of the kernel is $r^2/t > r$, and its denominator zero lies on the removed slit. Thus the kernel is harmonic away from the slit. Moreover,

$$|r^2 - tz|^2 - r^2|z - t|^2 = (r^2 - t^2)(r^2 - |z|^2).$$

This proves positivity inside the disk and zero boundary value. The integrated logarithmic singularity is continuous, including at the slit endpoints; this follows either by its elementary antiderivative or by splitting into a short interval about the singularity and a uniformly continuous complement. For $\xi \in [u, v]$, the positive numerator is at most r^2, so

$$U_r(\xi) \le \ell_0 \log r - \int_u^v \log|\xi - t|\, dt \le \ell_0[1 + \log(2r/\ell_0)].$$

Indeed the last logarithmic integral is maximal at the midpoint, where each half has length $\ell_0/2$. Positivity and the maximum principle on the slit disk now give $0 \le h_r \le 1$. At $z = 1$ all ratios in (9) exceed one, proving strict positivity. □

Theorem 4.2 (Quantitative operator transfer)**.** *Let K be a matrix-valued holomorphic function on a neighborhood of the closed disk $|z| \le r$. Suppose*

$$\|K(z)\|_{\mathrm{op}} \le M \quad (|z| \le r), \qquad \|K(t)\|_{\mathrm{op}} \le \varepsilon \quad (u \le t \le v), \qquad 0 < \varepsilon \le M.$$

Then, without a matrix-dimension prefactor,

$$\|K(1)\|_{\mathrm{op}} \le M^{1-\eta_r} \varepsilon^{\eta_r}. \tag{10}$$

In particular, (4) and Corollary 3.2, with $0 < u < v < \min(z_*, 1)$, imply $\|K_\lambda(1)\|_{\mathrm{op}} \le D_\lambda^{-a_0 \eta_r}$.

Proof. For any two unit vectors x, y, set $f(z) = x^* K(z) y$. If f is not identically zero, then

$$\log|f(z)| - \log M + \log(M/\varepsilon) h_r(z)$$

is subharmonic on Ω_r, with nonpositive boundary upper limits: on the circle $h_r = 0$, and on the slit use $\log|f| \le \log \varepsilon$, $h_r \le 1$. The boundary-limsup maximum principle gives the bound at 1. Zeros of f have value $-\infty$ and cause no exception. Taking the supremum over x, y proves (10). Identically zero coefficients satisfy it directly. If $\varepsilon = 0$, the identity theorem applied to every coefficient gives $K \equiv 0$; this is the only use of identity continuation needed for that limiting case. □

4.1 The gain relative to the source barrier

With $F(x) = x\log x - x$, direct integration gives

$$J_r = F(r^2-u) - F(r^2-v) - \ell_0 \log r - F(1-u) + F(1-v). \tag{11}$$

For a fixed slit $0 < u < v < 1$, differentiation under this finite integral is uniform as $r \downarrow 1$. Hence

$$\lim_{r\downarrow 1} \frac{\eta_r}{r-1} = \frac{2\log((1-u)/(1-v)) - \ell_0}{\ell_0[1+\log(2/\ell_0)]} > 0. \tag{12}$$

The integrand derivative at $r = 1$ is $2/(1-t) - 1 = (1+t)/(1-t) > 0$.

By comparison, the source uses $\int_u^v -\log|z-t|\,dt$, requiring $v < r-1$ to make it nonpositive on the outer circle. Take $u = c_1(r-1)$, $v = c_2(r-1)$, $0 < c_1 < c_2 < 1$. Its maximal slit value is $\ell_0[1+\log(2/\ell_0)]$, whereas its value at one is $\ell_0(u+v)/2 + O(\ell_0(r-1)^2)$. After normalization the endpoint saving has order $(r-1)/\log(1/(r-1))$, instead of the linear order in (12). This is a comparison of these explicit uniform-density barriers, not a claim of optimal harmonic measure. It keeps the real interval available independently of $r-1$. If z_* itself deteriorates in a varying family, that dependence must also be retained.

Corollary 4.3 (Dimensional majorant and error budgets)**.** *Suppose instead that $D \geq 1$, $a > 0$, $\|K(z)\| \leq CD^\sigma$ on the disk, $C \geq 1, \sigma \geq 0$, and $\|K(t)\| \leq D^{-a}$ on the slit. Then*

$$\|K(1)\| \leq C^{1-\eta_r} D^{-a\eta_r + \sigma(1-\eta_r)}.$$

A positive power saving requires $a\eta_r > \sigma(1-\eta_r)$. With disk bound M and $e_D \geq 0$, a slit bound $D^{-a} + e_D \leq M$ gives the transferred bound $M^{1-\eta_r}(D^{-a}+e_D)^{\eta_r}$.

Proof. Both assertions are substitutions in Theorem 4.2. □

The source has $\sigma = 0, C = 1$, which is decisive. A polynomial's degree, or holomorphy on a common set without a majorant, does not supply this budget. For instance $p_n(z) = z^n$ tends uniformly to zero on every $[u,v] \subset (0,1)$ but $p_n(1) = 1$; its disk norm r^n records exactly the missing cost.

Example 4.4 (A rationally certified endpoint exponent)**.** As a conditional calibration, suppose a family has disk bound one at $r = 5/4$ and interval bound D^{-a} throughout $[1/8, 1/4]$. These are hypothetical constants, not asserted values of the source's R, z_*. The integrand in J_r increases in t, so

$$\eta_r \geq \frac{\log(46/35)}{1+\log 20} > \frac{11}{162} > \frac{1}{16}.$$

For the strict inequality, $\log(46/35) \geq 2(46-35)/(46+35) = 22/81$, by integrating the geometric series for $\log((1+x)/(1-x))$. Also $e^3 > \sum_{j=0}^{8} 3^j/j! > 20$, so $\log 20 < 3$. Thus $\|K(1)\| \leq D^{-a/16}$ follows without a numerical logarithm oracle. The exact integral usually gives a better constant.

5 An explicit finite Fourier remainder

We record what the improved endpoint exponent does and does not imply for the full permutation law. Suppose the source's dimension bounds hold: for $k(\lambda) = \min(N-\lambda_1, N-\lambda_1')$,

$$D_\lambda \geq e^{-1}(N/k)^k \quad (1 \leq k \leq N/3), \qquad \log D_\lambda \geq c_{\dim} N \quad (k > N/3),$$

the latter for sufficiently large N, with fixed $c_{\dim} > 0$. There are at most $2 \cdot 2^k$ diagrams at level k, and at most 2^N diagrams in total. These are the source's hook and counting inputs, not conclusions from the line-gap comparison.

Proposition 5.1 (Reciprocal-dimension remainder)**.** *Assume the above bounds and* $\|K_\lambda(1)\| \le D_\lambda^{-g}$*, with the sign coefficient zero. Let* $w \ge 1$ *be an integer, and set*

$$\beta = 2gw - 2 > \max\{\log 2/\log 3, \log 2/c_{\dim}\}, \quad q_N = 2N^{-\beta/2} < 1, \quad \tau_\beta = 2\,3^{-\beta} < 1.$$

For all sufficiently large N*, the law* $\mu_w = B_N^{*w}$ *satisfies*

$$4\,\|\mu_w - U_N\|_{\mathrm{TV}}^2 \le 2e^\beta \left[2N^{-\beta} + \frac{4N^{-\beta}}{1-q_N} + \frac{\tau_\beta^{\lceil\sqrt{N}\rceil}}{1-\tau_\beta}\right] + e^{-(\beta c_{\dim} - \log 2)N}. \tag{13}$$

For fixed admissible β*, this is* $\|\mu_w - U_N\|_{\mathrm{TV}} = O_\beta(N^{-\beta/2})$.

Proof. Finite-group matrix Plancherel and Cauchy–Schwarz give

$$4\,\|\mu_w - U_N\|_{\mathrm{TV}}^2 \le \sum_{\lambda \notin \{(N),(1^N)\}} D_\lambda^{-\beta}.$$

The sign representation is omitted because one independent fair switch already makes its expectation zero. The trivial representation must be subtracted, rather than estimated by a strict contraction. The source dimension/count bounds make the near-row/column contribution at most

$$2e^\beta \sum_{k=1}^{\lfloor N/3\rfloor} [2(k/N)^\beta]^k.$$

The $k = 1$ term is $2N^{-\beta}$. For $2 \le k < \lceil\sqrt{N}\rceil$, the base is at most $2N^{-\beta/2} = q_N$, so the sum is at most $q_N^2/(1-q_N)$. For the remaining $k \le N/3$, the base is at most τ_β; extend the geometric sum to infinity. The far diagrams contribute at most the last exponential in (13). For fixed β, the geometric denominators are bounded for large N, and both exponential terms are negligible relative to $N^{-\beta}$. Taking square roots proves the assertion. □

This turns the source's dominated-convergence conclusion into an explicit remainder, conditional on exactly the same dimension and operator inputs. It does not compute w for the source: the values of R, q, z_*, and an effective onset for the dimension bound are still required. A larger certified g can reduce a sufficient integer w when those other constants are fixed, but no numerical source mixing-time improvement is measured here.

5.1 Clocks, order, and deterministic starts

Write $d = b\ell + t$, $0 \le t < \ell$, for $d \ge \ell = \log_2 R$. Block the bits consecutively into one block of $\ell + t$ bits and $b - 1$ blocks of ℓ bits. Each block size is in $[R, R^2]$. At $z = 1$, independent lines inside a block may be interleaved by bit direction; the resulting ordered sweep is the original B_N. No interchange of dependent successive block operations is intended.

A physical Thorp step is a deterministic bit rotation together with independent binary switches in the corresponding pair direction. Conjugate the successive switches by the rotations: over d steps the directions run once through the bits and the total rotation is the identity. Thus w full sweeps mean wd physical steps. Translation by a deterministic initial permutation preserves distance from uniform, so the same upper bound applies to every initial deck. There are at most $2^{tN/2}$ histories after t physical steps; comparison with $N!$ uniform states yields

$$\|\mu_t - U_N\|_{\mathrm{TV}} \ge 1 - \frac{2^{tN/2}}{N!}.$$

In particular a $1/4$ mixing threshold requires $t \ge (2/N)\log_2(3N!/4) = 2d - O(1)$. This is the source's support argument, included to pin the time unit; it is not counted as a new local contribution.

6 Proof architecture and retained obligations

All seven scientific source modules were freshly read in their entirety. The map below separates their logical functions from the selected deductions established here.

Introduction and preliminaries. The introduction fixes the physical shuffle, exact sweep identification, conditioned theorem and support lower bound. The preliminaries fix unnormalized Schatten norms, branching and signed occurrences, and the near-row/column and bulk Specht-dimension estimates. The representation-theoretic external facts remain classical inputs.

Prescribed paths. Feasibility is layerwise injectivity of coordinate-respecting paths. It makes each line's partial assignment injective and gives positive conditioning probability near $z = 0$. The compatibility potential and its exact increments are essential for later cancellation. No law conditioned on a typical outside environment can simply be substituted for this all-feasible-path statement.

Sparse representations. The source alternates placement probabilities over subsets of moved particles. Proper subsets vanish on the appropriate representation level. The many-coordinate argument uses sharing forests and one-card cancellation. The bounded-coordinate argument uses inverse-gamma variables on individual lines and an exact moment formula

$$W = m/Z, \quad Z \sim \mathrm{Gamma}(m - h + 1, 1), \qquad \mathbb{E}W^v = \frac{m^v}{(m-h)_v}$$

where the integer moments exist in the stated free-slot range. Factors on a shared line use the same variable W, not independent copies. Line grouping precedes taking moments. Colliding paths inside the alternating expression are not casually removed. Heavy-line and slot-sharing events require their separate combinatorial estimates. The eventual small interval in z is obtained after fixing the other constants; its size is not supplied by a generic analytic continuation principle.

Signed tensor densities. The source uses an even trace-one density on a graded $p|p$-dimensional space and signed tensor actions. The hook projector is dominated by a density mixture with the precise multiplicity normalization and an $O(p^2 \log(s+1))$ cost. Replacing it by a rank bound or independent parity choices would change that estimate. The row–column overlap proof depends on positivity and actual trace normalization, not merely the existence of a Gram matrix.

Simultaneous induction. The source fixes the sparse exponent and moment order before choosing large R, and only then a common small z_*. Balanced block splitting retains all feasible hole configurations. The hook overlap has the schematic exponent

$$F_R + F_C - F_\gamma + h' + O((s_1 + s_2)p^2 \log s).$$

Child moments cancel the entropy sum with the needed fractional powers. The perturbed coefficients provide reserves to absorb errors. Outside the hook, restriction to k removed boxes creates placement fibers, and cyclic trace Hölder estimates combine their probability factors with the compatibility-potential increment. The final fractional transition sum is controlled using both row and column stochasticity. A fixed-H or fixed-λ proof would not suffice for these applications of the induction hypothesis. These are imported source mechanisms; this report does not replace their simultaneous proof with tests.

Analytic transfer. The source explicitly restricts complex parameters to the unconditioned polynomial. Its invariant/complement splitting supplies a disk majorant uniform in representation dimension and grid size. Its logarithmic potential then proves a positive endpoint power, followed by Fourier summation and exact clocks. This part is mathematically stronger than the catalogue's initial warning about identity continuation alone. We retain the source argument and refine its constants by Theorem 3.1, Theorem 4.2, and Proposition 5.1. No source correction is alleged in this report.

7 Fresh original Ram comparisons

The two confidential originals cited here were inspected locally in bounded literal displays; they were not supplied to a search engine. Their role is analytic discipline and exact domain transport, not a pre-existing proof of shuffle mixing.

7.1 Original analytic-continuation theorem

The original TR018 proof [5], lines 1135–1248, specifies a connected order domain with fixed branches and histories, together with locally uniform integrable majorants. It distinguishes holomorphic moments from meromorphic inverses, and uniqueness by the identity theorem from stable reconstruction. In particular, reciprocal Gamma factors do not by themselves prevent a moment integral or inverse from developing a pole.

The direct analogue here is concrete. For the unconditioned sweep, finite sums and products prove entire dependence in z; no integration/interchange theorem is required. For conditioned laws, Proposition 2.1 identifies the actual denominator. For quantitative transfer, a common holomorphic domain must be supplemented by the disk norm. Corollary 4.3 quantifies the loss if that majorant grows with dimension. This is the additional bound that a formal analytic identity cannot provide.

7.2 Original kernel-coordinate equivalence

The original TR117 proof [6], lines 210–325, starts with an ordinary absolutely integrable kernel expression

$$g(x) = \int_r^s h'(x,\xi) f(\xi)\, d\xi = \int_{x-s}^{x-r} h(x-\alpha,\alpha) f(x-\alpha)\, d\alpha,$$

where $h(x,\alpha) = h'(x+\alpha,x)$. Its inverse coordinate relation is $h'(x,\alpha) = h(\alpha, x-\alpha)$. The proof is the substitution $\alpha = x - \xi$, including reversal of bounds; this is an inverse *kernel-coordinate map*, not an inverse of an arbitrary integral operator. Differentiation or singular limits need further hypotheses.

Applied as a proof discipline, the analogy is with preserving free-slot identifications, product order and feasible events, rather than treating an algebraically equivalent matrix expression as a new stochastic law. It also prevents replacing the subharmonic comparison by a numerical interpolation of finitely many matrix values. Neither private original is claimed to contain the cube-word constant or the fixed-slit endpoint exponent derived here.

8 Current public status

The public refresh was performed on October 9, 2026 using public titles and URLs only. Morris's arXiv abstract, submitted December 14, 2009, and the Cambridge publisher record identify the

published $O(d^3)$ bound for 2^d cards [2]. The journal issue is January 2013, volume 22, pages 118–132; online publication was December 7, 2012. Its full proof was not independently audited here.

The publisher's family-238 formalization scope page [3] states a binary representation contraction and a separate feasible conditional moment estimate for this manuscript, and explicitly excludes physical-shuffle law identification from those selected statements. Its neighboring trace-smoothing statement concerns a different companion. This is publisher scope metadata, not a Lean execution or formal-proof audit performed for this report. We do not use it to remove any imported hypothesis.

The boundary-limsup maximum principle used in Theorem 4.2 is classical; the source cites Demailly, Chapter I, Section 4.C.1, Theorem (4.14) [4]. The primary-site indexed theorem passage was checked; a direct PDF fetch timed out. No complete external-book read is claimed. The Green kernel and its normalization are proved explicitly here, so the quantitative constants do not depend on an unexamined numerical harmonic-measure computation.

9 Advantages, costs, and further work

Interface	Retained source statement	Local increment and limitation
Line contraction	A finite maximum of strict irreducible contractions is below one.	An explicit word-energy deficit avoids irreducible enumeration, but is very conservative.
Complex transfer	A uniform disk and a shrinking logarithmic slit give a positive power.	A Green slit need not shrink with $r-1$; its explicit exponent still depends on z_*.
Full-law summation	Reciprocal dimensions tend to zero by domination.	An explicit geometric remainder is obtained, conditional on the same dimension and operator bounds.
Conditional laws	Feasible-path moments near $z = 0$.	Exact poles and a zero-probability endpoint example identify what cannot be transferred as a physical conditional law.

The gain is an explicit quantitative interface, not a shorter proof of the difficult simultaneous theorem. The line deficit can be poor when the allowed block size is large; improving it with a proven comparison or a rigorously enclosed finite spectrum is possible, but a sampled set of irreducibles is insufficient. A better slit density or the actual harmonic measure may improve η_r, subject to a certified boundary comparison. Optimizing r, the slit, and the source parameters jointly requires the dependence of z_* and q on R; the fixed-slit asymptotic alone cannot optimize that dependence.

A perturbative implementation should keep separate errors for the line law, the conditional denominators, and the real-interval norm bound. A holomorphic error with a proven disk norm can enter Corollary 4.3. Arbitrary pointwise numerical errors on a grid cannot. Extension to nonbinary blocks would require a generating family and a new word-length/projection estimate, as well as a replacement for the source's exact binary law identification.

10 Evidence and conclusion

Fresh literal reading covers all 11 cached text files, 1,807 lines: the seven scientific sections comprise 1,592 lines, and the main, README, bibliography and trajectory figure source comprise 215. The entire scientific source is thus read, but the induction and its external representation-theory inputs are not independently certified. Private reading covers 230 lines: TR018 lines 1135–1248 of 2,244,

and TR117 lines 210–325 of 1,794. Their explicit unread complements and pre/post SHA-256 values are in the owned read ledger. No unlisted inclusion, prior assignment, parent check, or corpus map is counted as an original-proof read.

The owned executable checks enumerate small switch laws and conditioning events, test rational radius and word identities, and verify exact finite inequalities behind the calibration and Fourier sum. Separate high-precision diagnostics compare the Green integral with its antiderivative and boundary formulas. Counts distinguish parameter cases from assertions and numerical diagnostics from exact checks: 1,192 exact cases passed 2,623 assertions, and 17 65-digit numerical cases passed 139 diagnostic assertions. The exact groups include all 432 selected conditioned four-slot laws, 651 cube transposition words, and 48 regular-S_4 energy fixtures. They supplement the analytic proofs; no finite set verifies the global source theorem. The final check receipt and freeze bind the code, results, report, read inputs and rendered-page evidence.

The final mathematical boundary is sharp: an unconditioned polynomial with a dimension-uniform disk bound admits the explicit Green transfer; a merely analytically continuable conditioned expression need not. The source already respects this distinction. The present contribution is to make its line radius, endpoint exponent and Fourier remainder more explicit while leaving the simultaneous probabilistic and representation-theoretic obligations visible.

References

[1] OpenAI, *Conditional coordinate sweeps and analytic transfer*, September 26, 2026. Pinned local manuscript, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory. All seven scientific modules read; global simultaneous moment theorem used as an imported input.

[2] B. Morris, *Improved mixing time bounds for the Thorp shuffle*, Combinatorics, Probability and Computing **22** (2013), 118–132. doi:10.1017/S0963548312000478; arXiv:0912.2759. Abstract and publisher metadata checked, not a full-proof audit.

[3] OpenAI, *Optimal logarithmic mixing of the Thorp shuffle*, family-238 formalization scope documentation, accessed October 9, 2026. Publisher description only; no Lean execution performed here.

[4] J.-P. Demailly, *Complex Analytic and Differential Geometry*, Chapter I, Section 4.C.1, Theorem (4.14). Author-hosted text. Classical subharmonic maximum principle; scoped access described in Section 8.

[5] M. Subbarao, original RamFoundationTRs manuscript `TR-rev-1-old-tr018.tex`, analytic-continuation theorem and interpolation corollary, freshly read lines 1135–1248. Confidential local original; not publicly uploaded.

[6] M. Subbarao, original RamFoundationTRs manuscript `TR-rev-1-old-tr117.tex`, domain assumptions and global/Ram kernel equivalence proof, freshly read lines 210–325. Confidential local original; not publicly uploaded.